Mechanisms of Attack: Chemical Attack in Concrete

Chemical Action

When we are dealing with durability, chemical attack which results in volume change, cracking and consequent deterioration of concrete become amajor cause of concern

Types of Chemical attack

- Sulphate attack
- Alkali aggregate reaction
- Chloride ion attack - Corrosion
- Carbonation
- Acid Attack
- Effect on concrete in Seawater

CHLORIDE ATTACK

- Chloride attack is particularly important because it primarily causes corrosion of reinforcement.
- Statistics have indicated that over 40 per cent of failure of structures is due to corrosion of reinforcement.

CHLORIDE ATTACK

- Due to high alkalinity of concrete a protective oxide film is present on the surface of steel reinforcement.
- This protective passivity layer can be lost due to carbonation and chloride in presence of H_2O and O_2.

CHLORIDE ATTACK

- Chloride enters the concrete from cement, water, and aggregate and sometimes from admixtures.

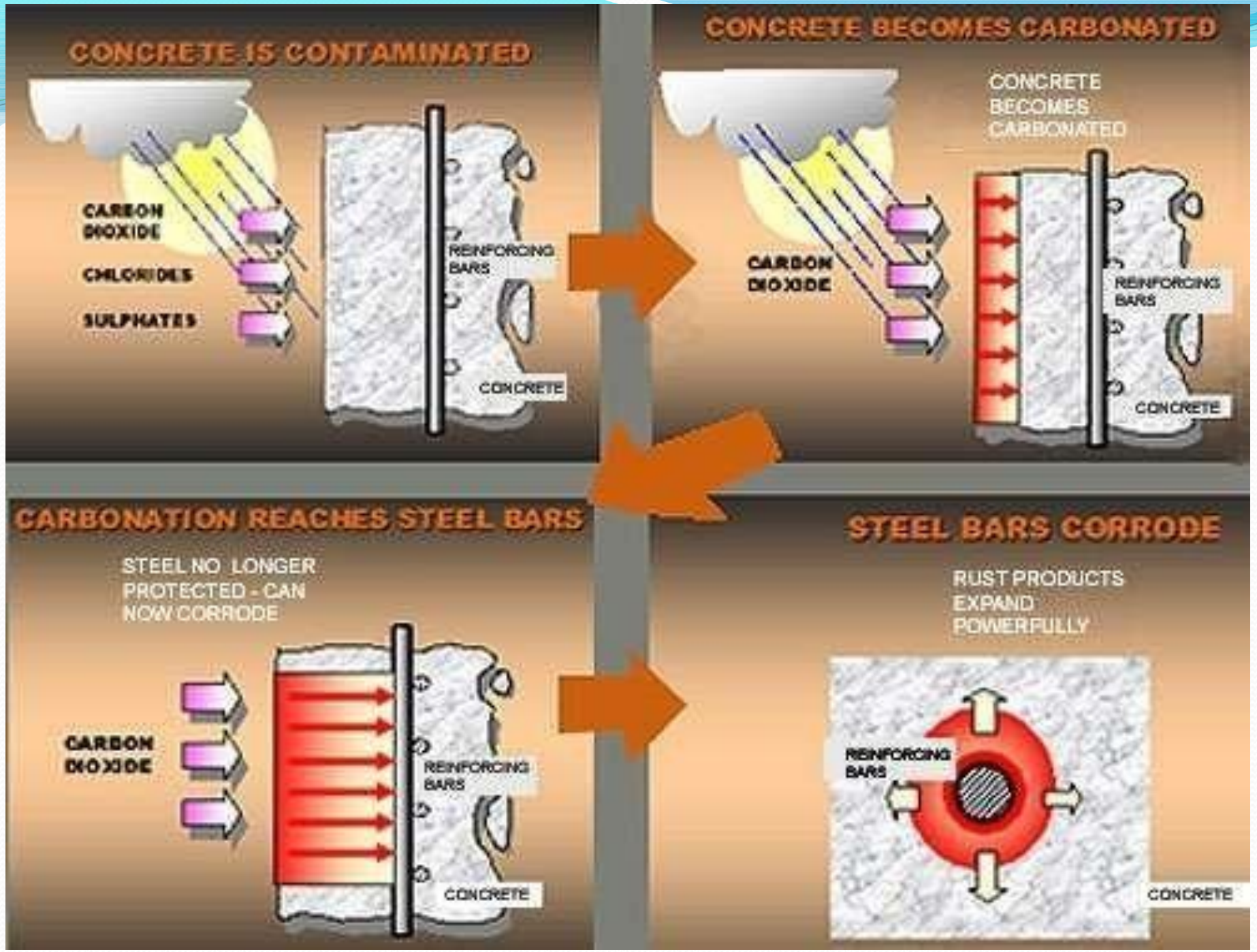
According to IS 456-2000

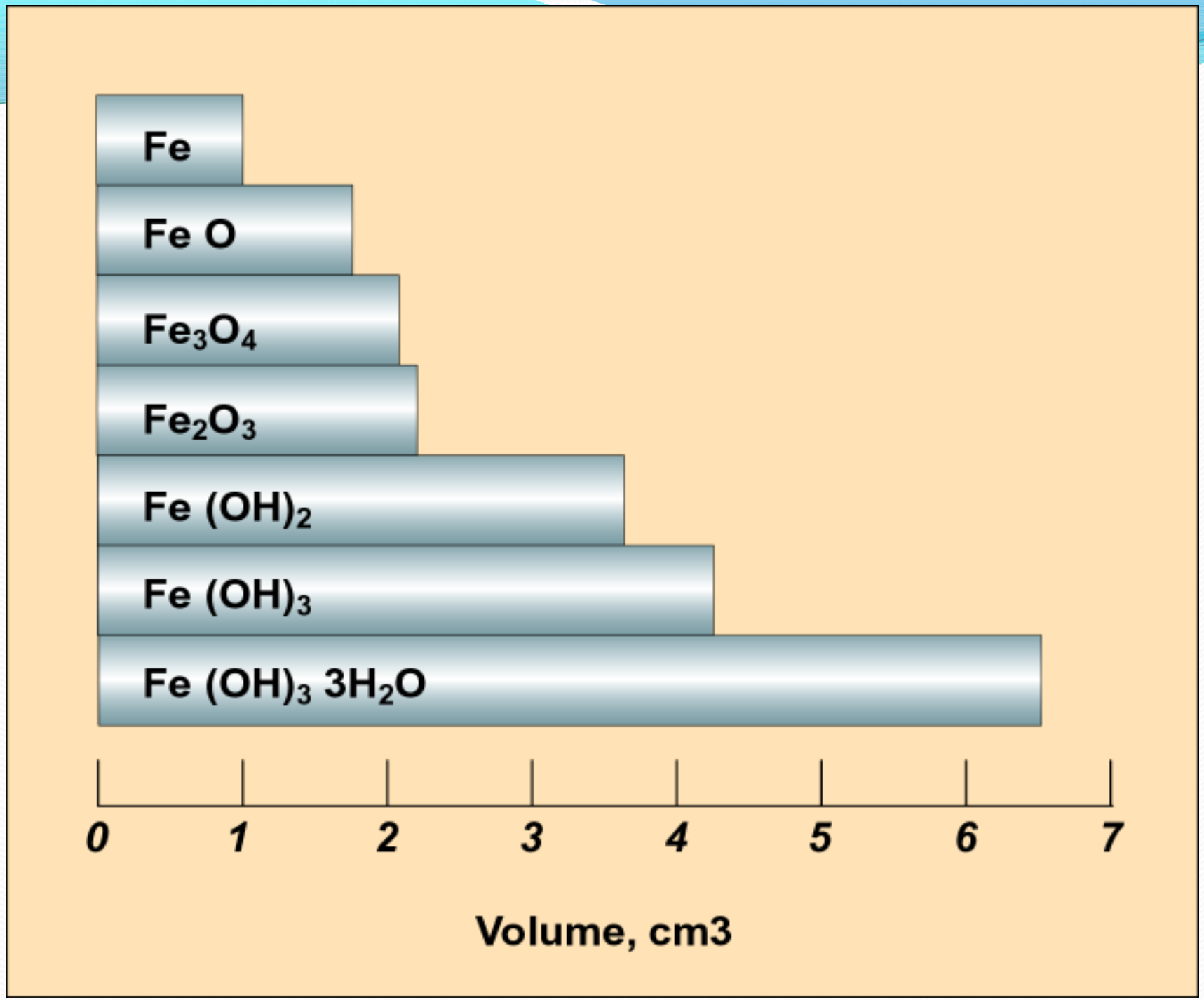
S.No.	Type of Use of Concrete	Maximum Total acid soluble chloride content. Expressed as kg/m^3 of concrete
1	Concrete containing metal and steam cured at elevated temperature and prestressed concrete.	0.4
2	Reinforced concrete or plain concrete containing embedded metal.	0.6
3	Concrete not containing embedded metal or any material requiring protection from chloride.	3.0

CHLORIDE ATTACK

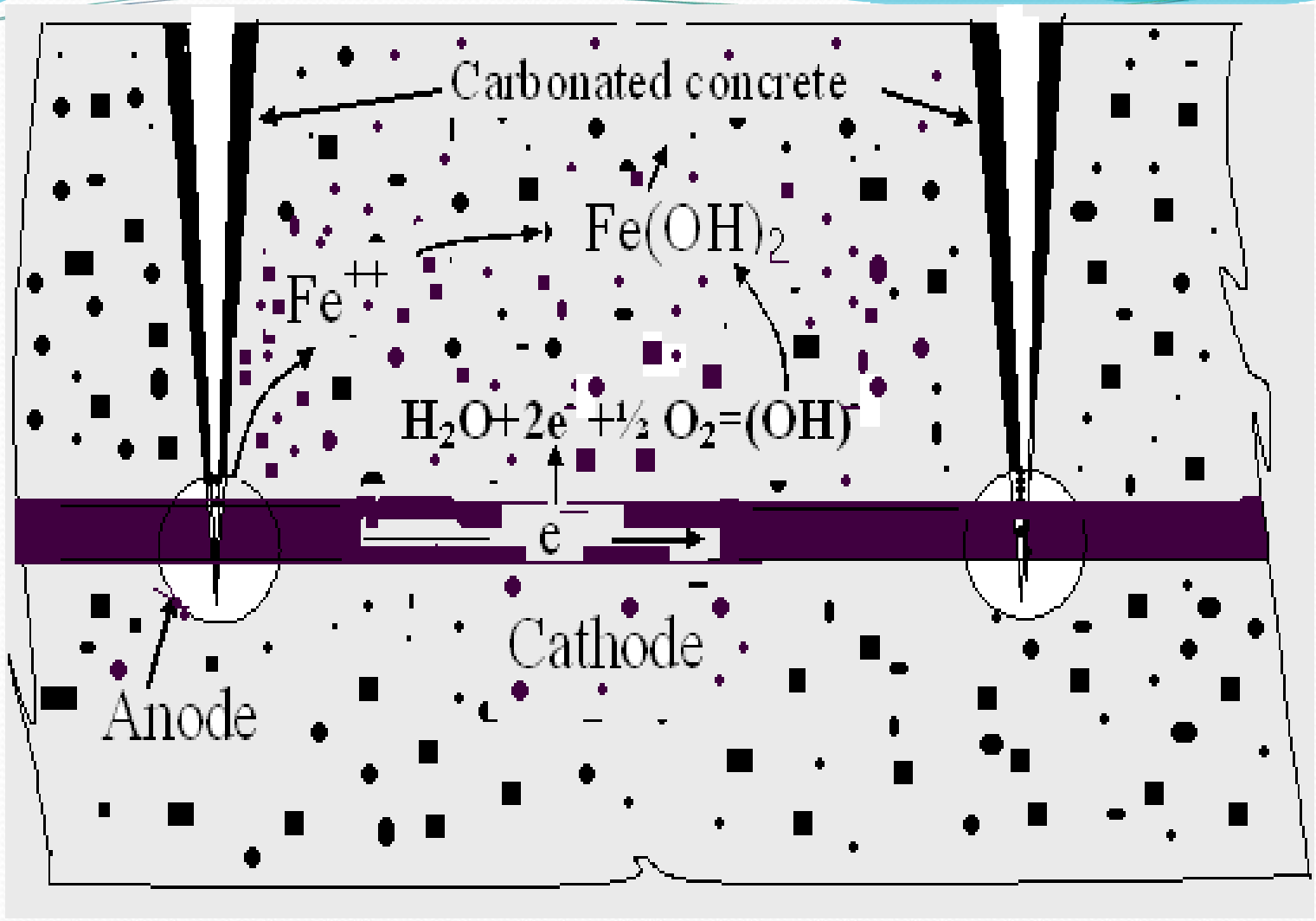
Prevention measures:

- Use supplementary cementitious materials to reduce permeability
- Increasing the concrete cover over the steel
- use of corrosion inhibiting admixtures
- epoxy-coated reinforcing steel, surface treatments, concrete overlays, and cathodic protection

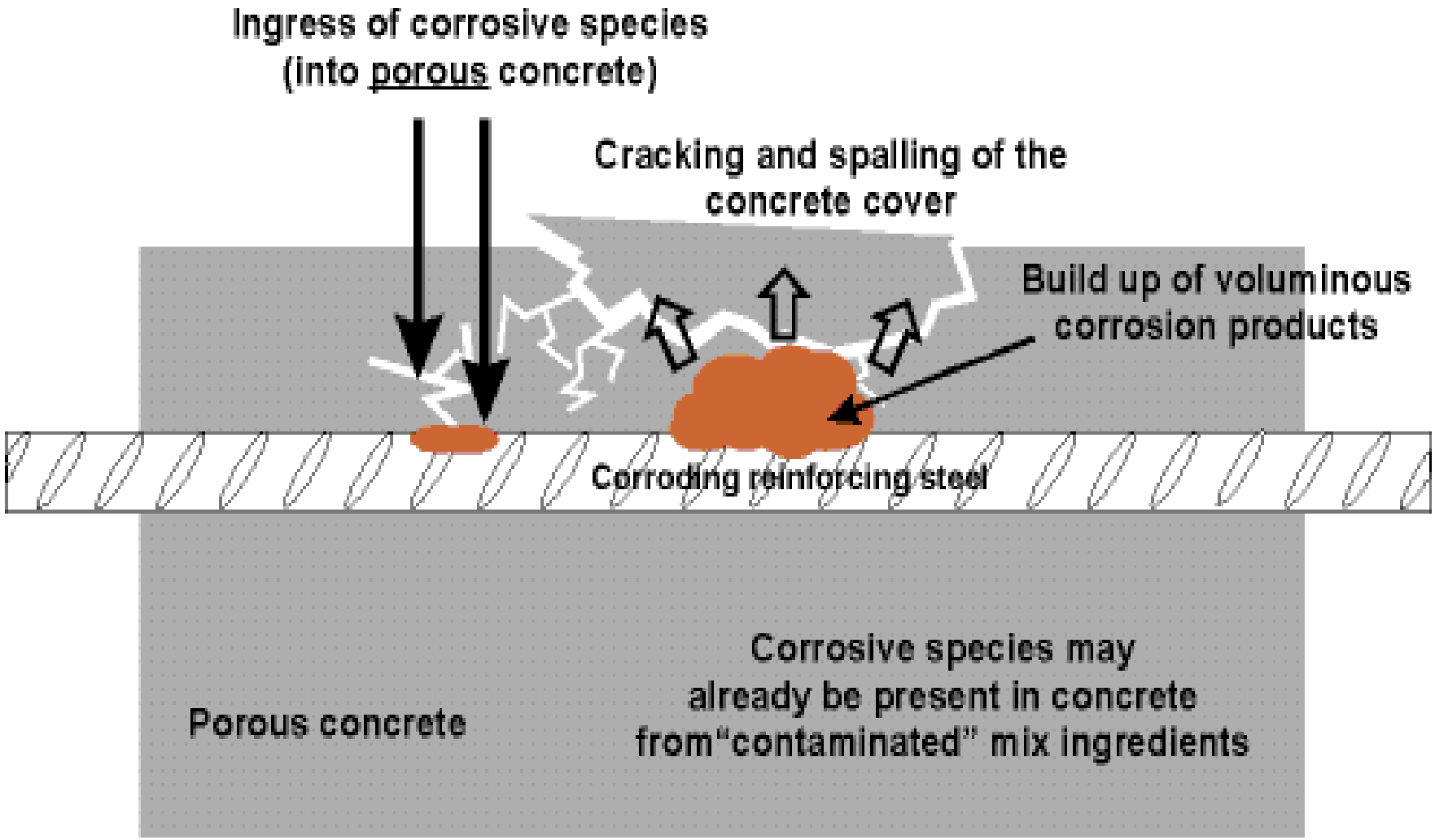

CONCRETE IS CONTAMINATED
CARBON DIOXIDE
CHLORIDES
SULPHATES
REINFORCING BARS
CONCRETE
CONCRETE BECOMES CARBONATED
CONCRETE BECOMES CARBONATED
CARBON DIOXIDE
REINFORCING BARS
CONCRETE
CARBONATION REACHES STEEL BARS
STEEL NO LONGER PROTECTED - CAN NOW CORRODE
CARBON DIOXIDE
REINFORCING BARS
CONCRETE
STEEL BARS CORRODE
RUST PRODUCTS EXPAND POWERFULLY
REINFORCING BARS
CONCRETE


Fe
Fe O
Fe_3O_4
Fe_2O_3
Fe $(OH)_2$
Fe $(OH)_3$
Fe $(OH)_3$ $3H_2O$
0
1
2
3
4
5
6
7
Volume, cm3


1113



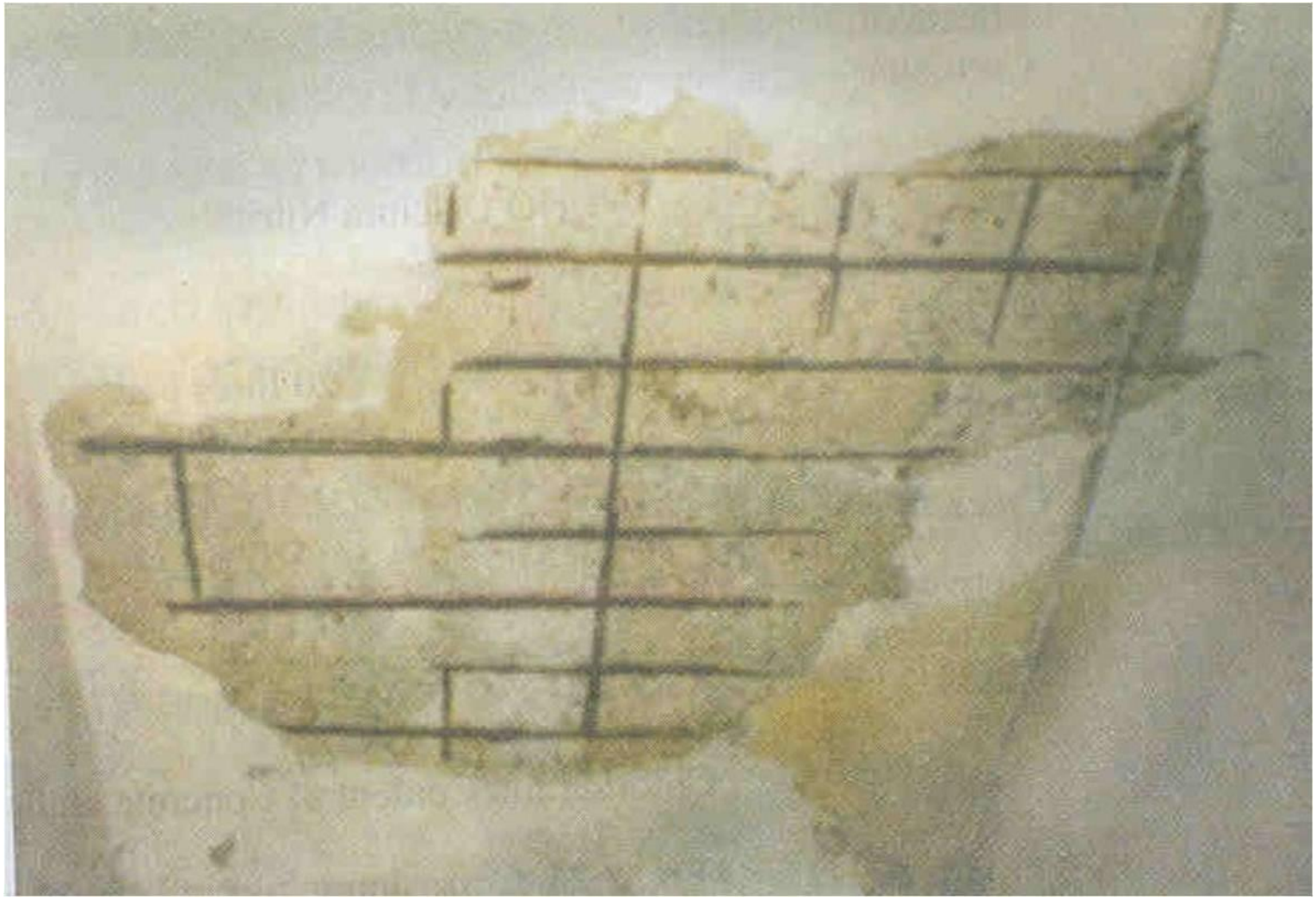
Mechanism of corrosion of steel in concrete


Ingress of corrosive species
(into porous concrete)
Cracking and spalling of the
concrete cover
Build up of voluminous
corrosion products
Corroding reinforcing steel
Porous concrete
Corrosive species may
already be present in concrete
from"contaminated" mix ingredients

Main Reasons of Corrosion

- Presence of chloride ions – from water for mixing as well as curing, from aggregates, other ingredients, from the service environment.
- Poor quality of concrete – high permeability, no control on water/cement ratio.
- Low pH value of concrete – mainly due to carbonation,
- Inadequate cover thickness.

Example of Delamination of Concrete cover



Crackformed due to bursting pressureon account of rusting of reinforcements



Prevention Measures

- Design Constraints
- Role of Materials
- Protective Coatings
- Cathodic Protection
- Environmental Modification
- Passivation

Design Constraints

- Design for proper drainage, less crevices, less stress, less water retention.
- Design for allowances.



Role of Materials

- Use of pure metals: eg: Aluminium, Titanium ,Gold, Platinum etc....
- Use of corrosion resistant alloys. eg:Duralumin, Stainless steel, Brass, Bronze etc....
- Fulfill required functionality in terms of strength and corrosion resistance.

Protective coatings

- Metal coating: Surface of a metal(base metal) is coated with another metal(coat metal).
- Painting: A suitable top coat which excludes water and oxygen from the surface.
- Polishing:Finely finishing the surface.

Metal Coating

- Hot Dipping: Base metal is dipped or passed through molten coat metal oralloy.
  Eg:Galvanization
- Metal Spraying: Coat metal is sprayed on base metal.
- Cementation: Base metal is protected by alloying its surface with another metal.

Painting

- Oil Painting: Most widely used consist of pigments and thinners.
- Water Painting: A suspension of dispersible pigments in emulsion of water and small quantities of drying oils.
- Special Paints: Special chemicals are added in paints specific purpose. Eg: Heat resistant paints, water repellant paints, etc.





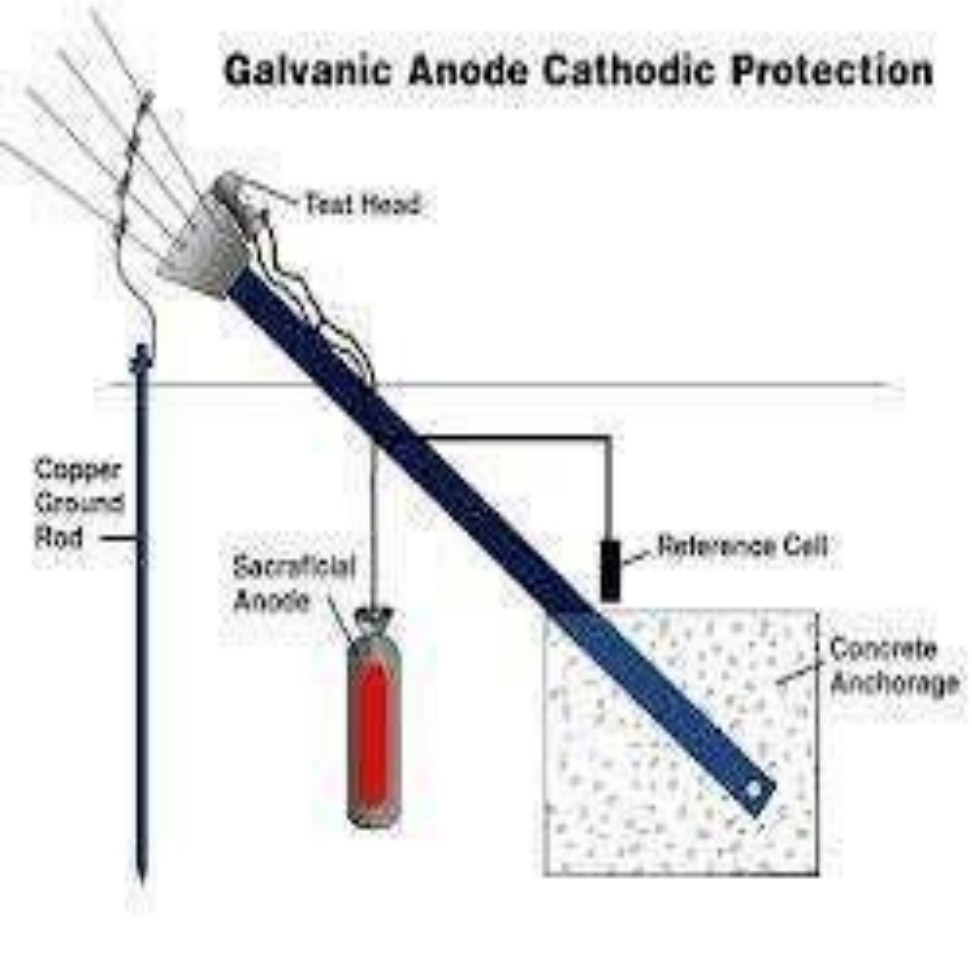
Cathodic Protection

- The metal to be protected is connected to an external anode and a D.C. current is passed so that all areas of the metal surface become cathodicand do not corrode.
- Cathodic protection can be achieved in two ways:

1. By Sacrificial Anodes.
2. By Impressed Current.

Sacrificial Anodes

- These anodes consist of the highly reactive metal like Mg.
- To protect pipes, rods of Mg buried near the pipes and connected to them by wires.
- Thus Galvanic cells with high E.M.F are setup with surface to be protected made cathodic.
- The cost of replacement of corroded anodic Mg is much less than replacing pipes.


Galvanic Anode Cathodic Protection
Test Head
Copper Ground Rod
Sacrificial Anode
Reference Cell
Concrete Anchorage


Sacrificial Anode
Water Main Pipe

Impressed Current Method

- The object to be protected is made cathode of an electrolytic cell by connecting it to negative terminal of a D.C source.
- This is done by rectifying an A.C line or using D.C source.
- The positive terminal is connected to scrap iron.

Environmental Modification

- Deaeration: Removal of dissolved oxygen. Eg: By adding chemicals like Sodium Sulphite.
- Dehumidification: Removal of moisture from air. Eg: By adding chemicals like Silica.
- Inhibitors: It is a chemical additive which reduce the rate of corrosion. Eg: Alkali, Amines, Orthophosphates, etc.

Case Study: Iron Pillar, Delhi

- The iron pillar of Delhi has not rusted fromthe last 16 centuries due to the following reasons:
  - Purity of its iron.
  - High Phosphorus content.
  - Low Sulphur and Magnesium.
  - Better forge welding.
  - Drier and uncontaminated atmospheric condition.



Case Study: Eiffel Tower

- The structure comprises 220000 metre square of surface (300 tonnes of wrought iron) that have to be maintained and repainted every 7 years, consuming50 tonnes of lead free paint.
- for renovation ,corroded areas are prepared before painting.
- All this is carried out by around 25 painters, works lasting for around 15 monthsand costing over 20 million francs.



Sulphate attack

- Sulphate attack denotes an increase in the volumeof cement paste in concrete or mortar due to chemical action between the products of hydration of cement and solution containing sulphate, and also sodium, magnesium and Cholorides.
- In hardened concrete, calcium aluminate hydrate (C-A-H) can react with sulphate salt from outside, product of reaction is calcium sulphoaluminate, which can cause an increase in volume up to 27%
- Rate of sulphate attack increases with asaturated sulphate solution.
- Asaturate solution of magnesium sulphate can cause serious damage to concrete with high w/c ratio.

Methods of Controlling Sulphate Attack

◊**Use of Sulphate resisting cement**: Use cement with low C_3A content is most effective method. So use sulphate resisting cement which contains less C3A.

◊**Quality Concrete** : A well designed, placed and compacted concrete exhibit higher resistance to sulphate attack.

◊**Use of air-entrainment**: Use of air-entrainment to the extend of about 6% has beneficial effect on sulphate resisting quality. This is probably due to reduction of segregation, improvement in workability, reduction in bleeding and in general reduction in permeability of concrete.

◊ **Use of pozzolona** : Use of pozzolanic materials reduce the permeability.

◊**High Pressure Steam Curing**: This improve the resistance of concrete to sulphate attack.

◊**Use of High Alumina Cement:** Use of High Alumina Cement improves the resistance of concrete to sulphate attack.



Spalling (Concrete flakes)



Spalling

Carbonation

- Carbonation of concrete is a process by which carbon dioxide from the air penetrates into concrete and reacts with calcium hydroxide to form calcium carbonates.
- In actual practice, CO_2 present in atmosphere permeates into concrete and carbonates the concrete and reduces the alkalinity of concrete.
- When all the $Ca(OH)_2$ has become carbonated, the pH value will reduce upto about 8.3. In such a low pH value, the protective layer gets destroyed and the steel is exposed to corrosion.

CARBONATION OF CONCRETE

It is a process by which CO_2 from the air penetrates into concrete and reacts with calcium hydroxide to form calcium carbonates in presence of water.

CH + CO2---------- CACO3 + WATER



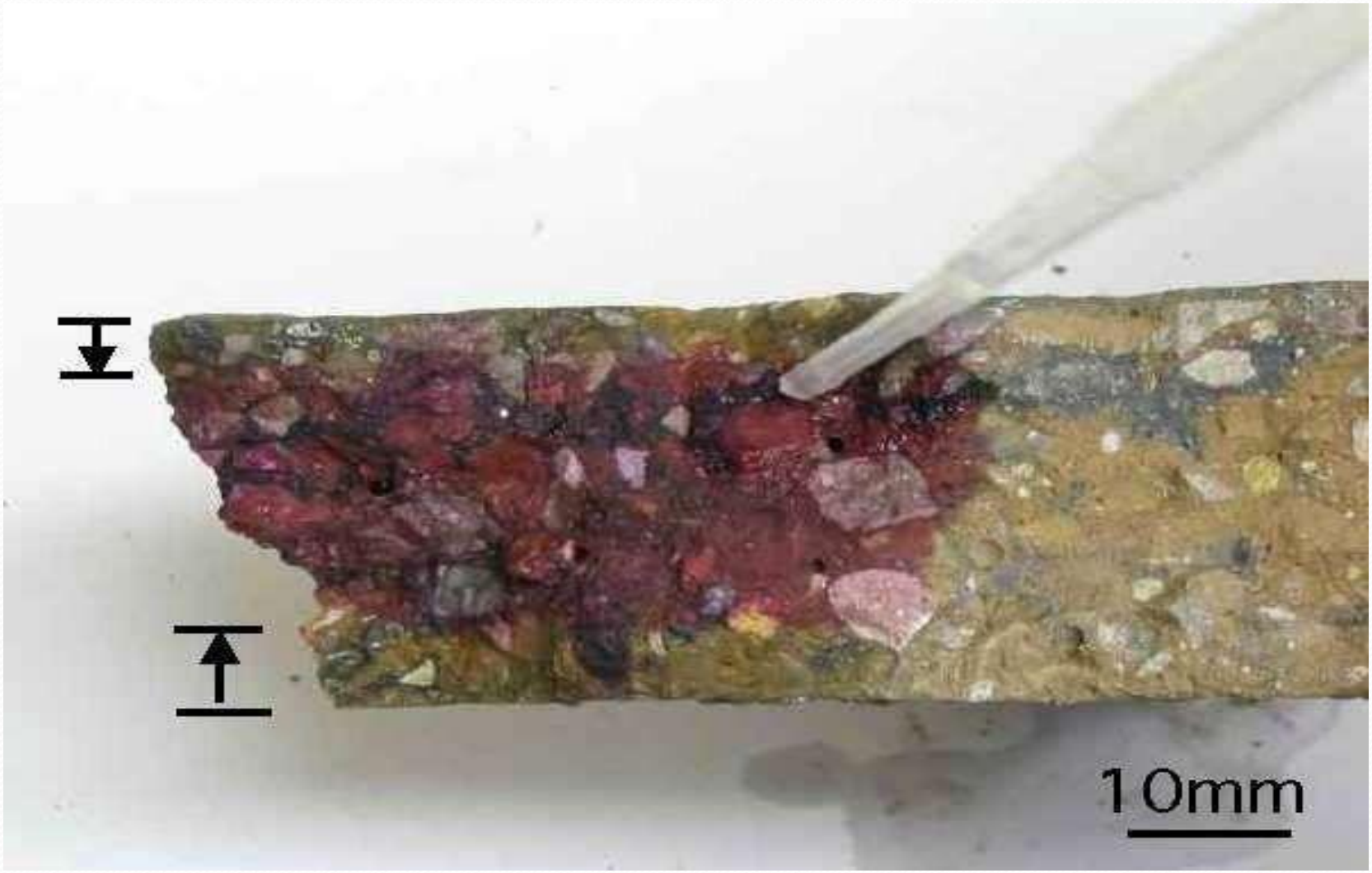
Carbonation in concrete

Measurement of Depth of Carbonation



Pink color indicates that Ca(OH)2 is unaffected by carbonation. The uncolored portion indicates that concreteis carbonated

CARBONATION OF CONCRETE



Alkali-Silica Reaction:
"The Cancer of Concrete"

Alkali - Aggregate Reaction

- Alkali from cement, reacts with reactive silica (of aggregates) to form alkali-silica gel of unlimited swelling type.
- The continuous growth of silica gel exerts osmotic pressure within the concrete.
- This manifests into cracking and bulging of concrete

Occurrence is due to :

1. High alkali content in cement (more than 0.6%)
2. Reactive silica in aggregate
3. Availability of moisture



What is ASR?

Alkali Silica Reaction (ASR)

Alkalis
+
Reactive Silica
+
Moisture

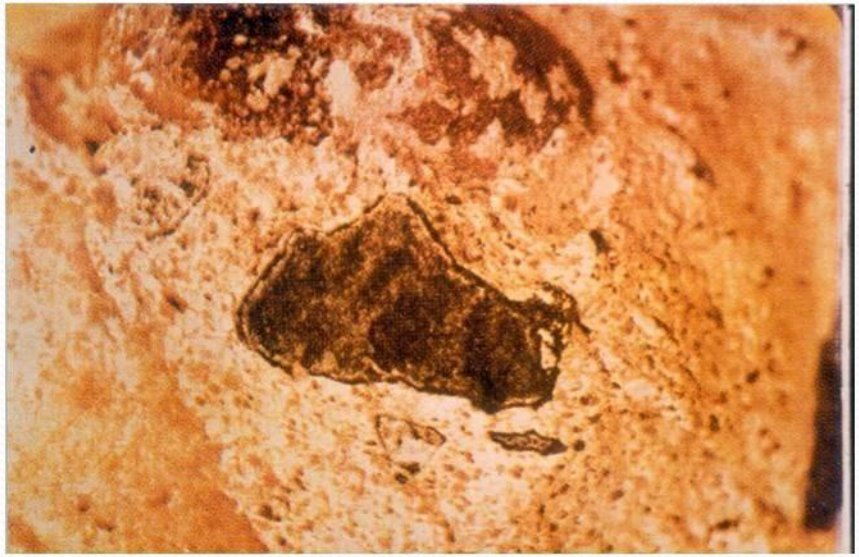


ASR Gel which expands



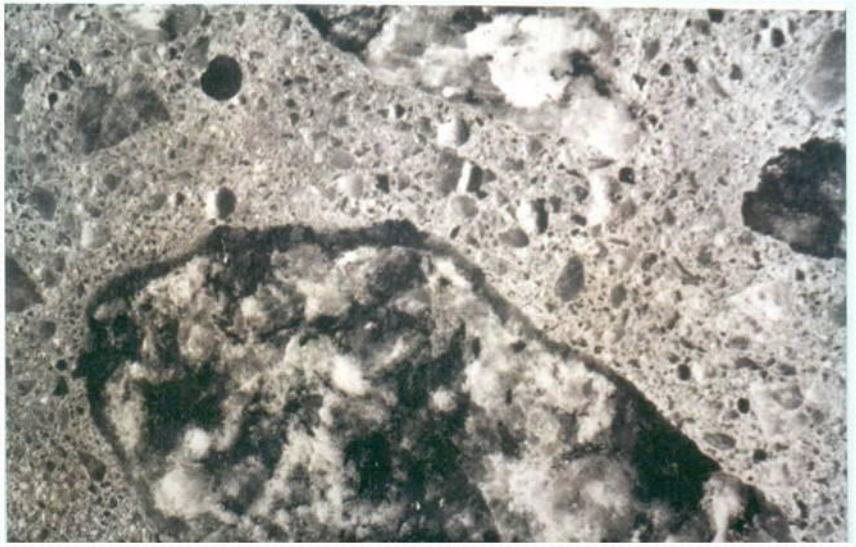
Concrete expansion and cracking

Alkali - Aggregate Reaction



Remedial Measures:

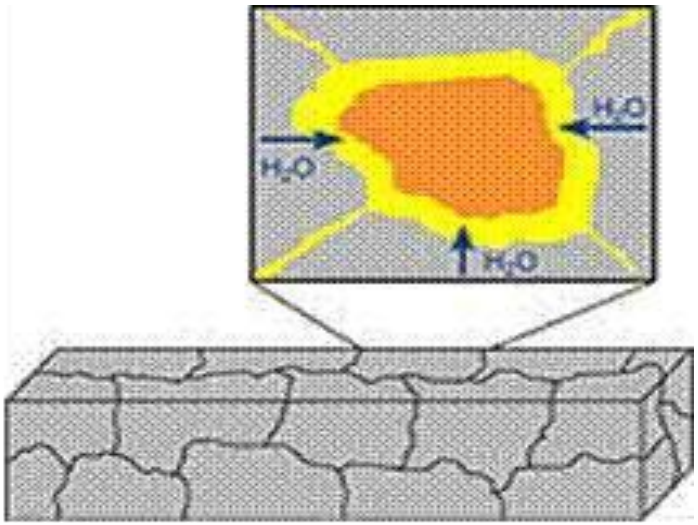
1. Use non-reactive aggregates from alternate sources
2. Use low-alkali cement
3. Reduce cement content in concrete
4. Use of pozzolana, slag or silica fume
5. Control on service condition, limiting degree of saturation of concrete

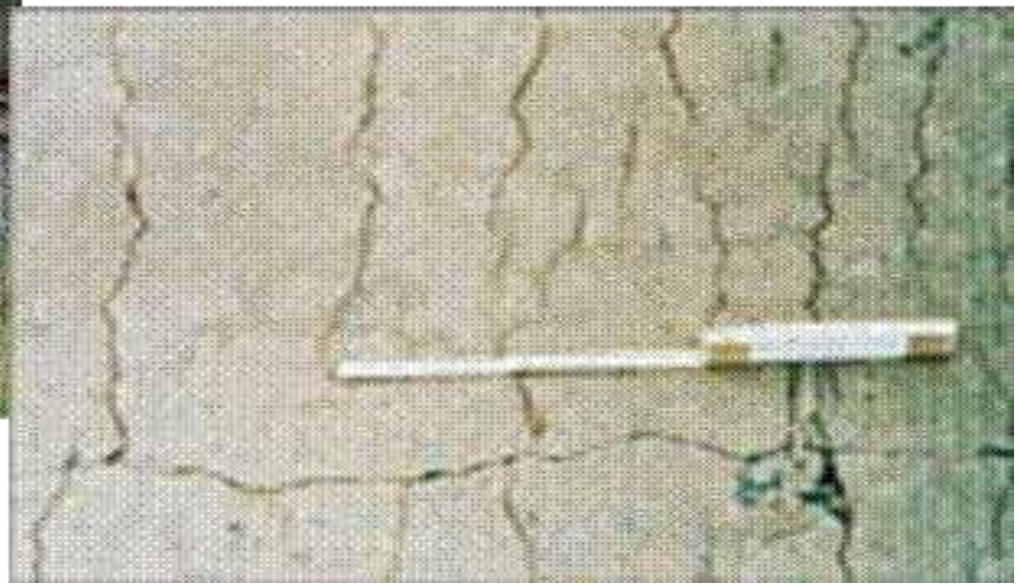


- Visual symptoms
  - network of cracks
  - Closed or spalled joints
  - Relative displacement
- cracking of concrete from alkali silica reactivity

Alkali silica reactions

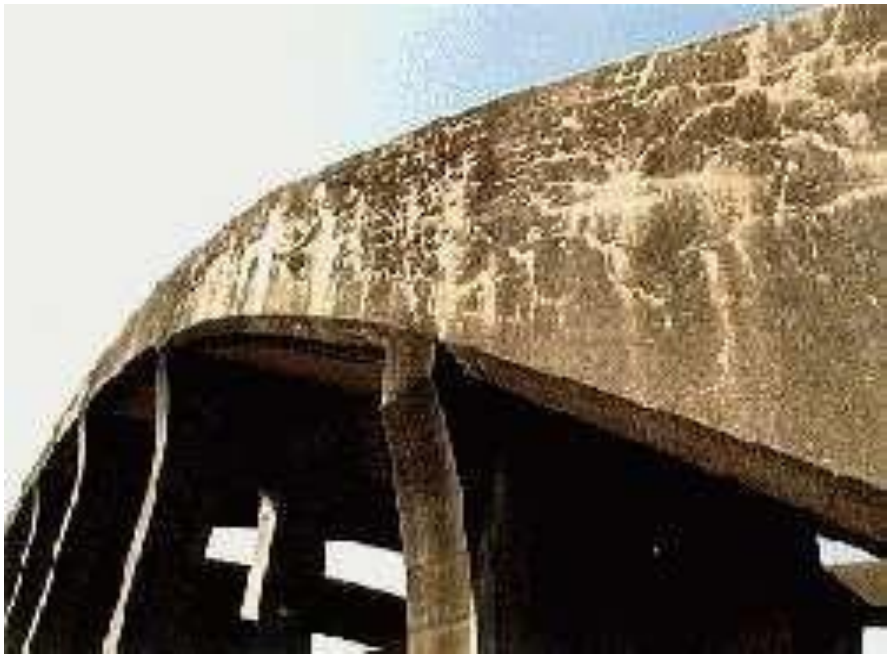






Concrete failure due to ASR





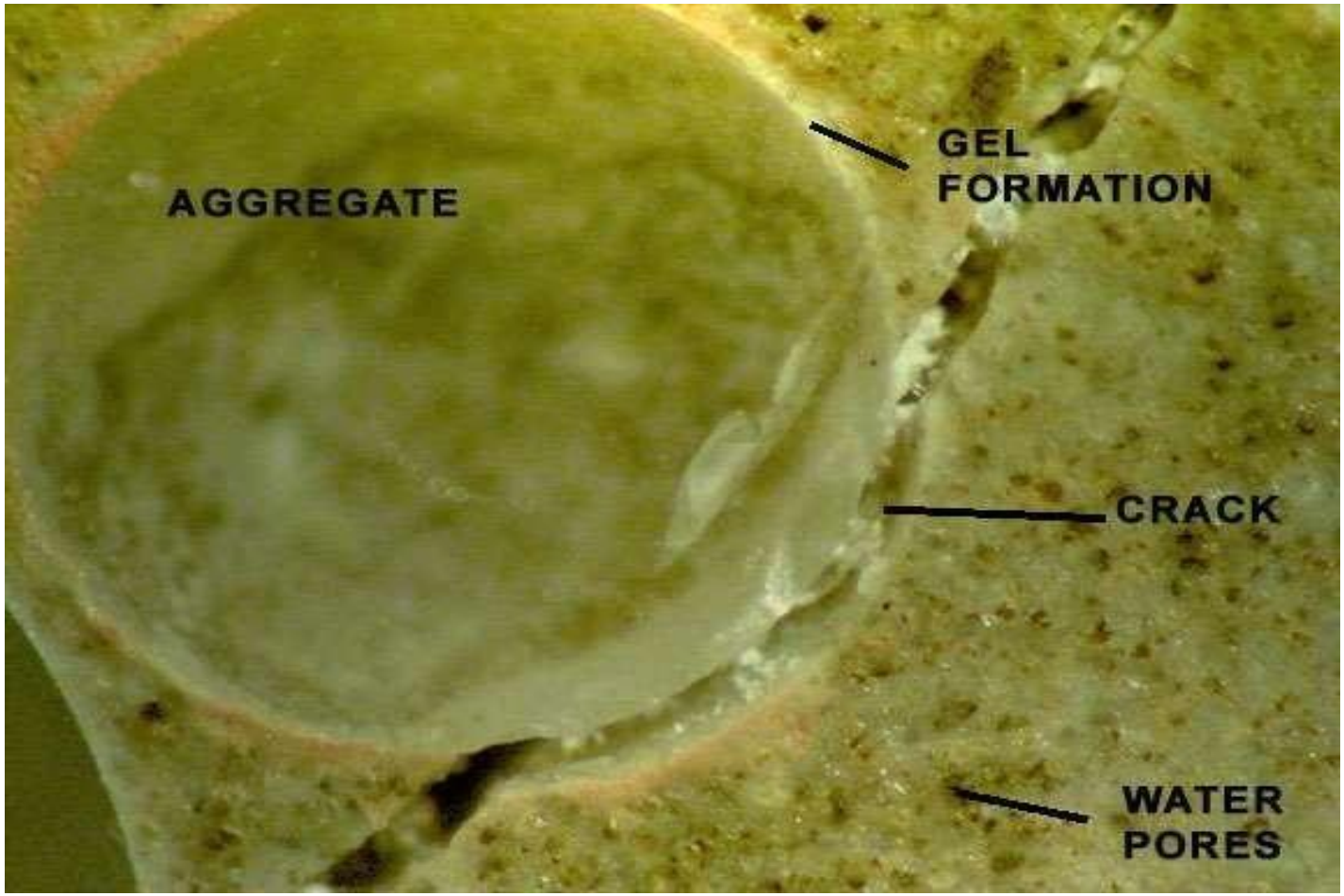




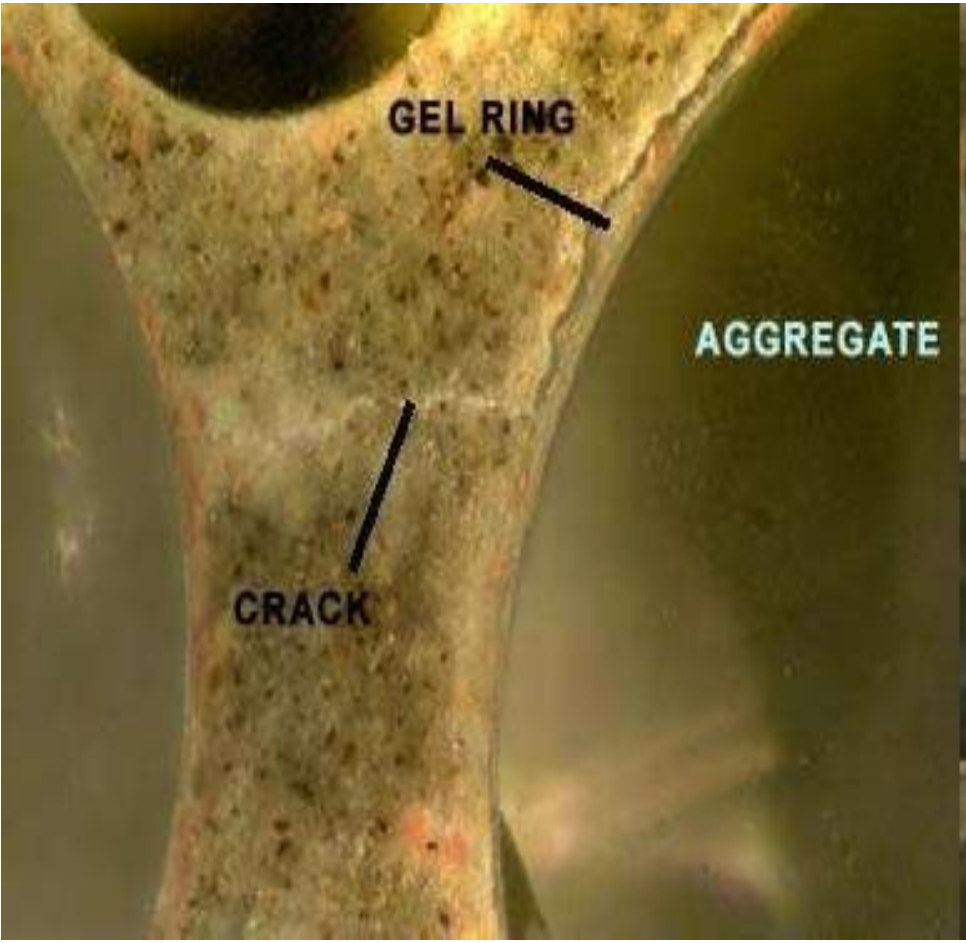
Why is it important to study ASR?

- *Concrete quality*
  - Loss of strength, stiffness, impermeability
  - Premature failure of concrete structures
- *Economic/Environmental impacts*
  - ASR decreases concrete service life
  - Reconstruction has both environmental and economic impacts. ex. cement production produces 7% of the world's CO_2 emissions (a greenhouse gas)

Images of ASR damage



Images of ASR damage contd…





Alkali-Silica Reaction



Controlling ASR

- Non reactive aggregates
- Supplementary cementing materials or blended cements
- Limit alkali loading
- Lithium based admixtures
- Limestone sweetening(30 % replacement reactive aggregates with crushed limestone)

Alkali carbonate reactions

- Influencing factors
  - Clay contents, or insoluble residue content, in the range of 5% to 25%
  - Calcite to dolomite ratio of approximately 1:1
  - Increase in the dolomite volume
  - Small size of the discrete dolomite crystals (rhombs) suspended in the clay matrix

Alkali carbonate reactions

- Controlling ACR
  - Selective quarrying to avoid reactive aggregates
  - Blend aggregates according to appendix in ASTM C 1105
  - Limit aggregates size to smallest practical

Acid attack

- Concrete is not fully resistant to acids depending upon the type and concentration of acid.
- Oxalic acid and phosphoric acids are harmless.
- The most vulnerable part of the cement hydrate Is Ca(OH)2, but C-S-H gel can also be attacked.
- Concrete can be attacked by liquids with pH value less than 6.5.

Acid attack

- But the attack is severe only at a pH value below 5.5.
- At a pH value below 4.5, the attack is very severe.
- Cement compounds are eventually broken down and leached away.
- If acids are able to reach the reinforcing steel through cracks corrosion can occur leading to further cracking